

1 gitinspector

gitinspector — statistical analysis tool for git repositories

Synopsis

gitinspector [OPTION]... [REPOSITORY]...

DESCRIPTION

Analyze and gather statistics about one or more git repositories. The default analysis shows general statistics per author, which can be complemented with a timeline analysis that shows the workload and activity of each author. Under normal operation, gitinspector filters the results to only show statistics about a number of given extensions and by default only includes source files in the statistical analysis.

Several output formats are supported, including plain text, HTML, JSON and XML. Every report names the branch that was analyzed in each repository; the checked out branch, or the commit when the repository has a detached HEAD.

OPTIONS

List information about the repository in REPOSITORY. If no repository is specified, the current directory is used. If multiple repositories are given, information will be merged into a unified statistical report.

Mandatory arguments to long options are mandatory for short options too. Boolean arguments can only be given to long options.

-f, --file-types=EXTENSIONS

A comma separated list of file extensions to include when computing statistics. The default extensions used are: java,c,cc,cpp,h,hh,*. Specifying a single * asterisk character includes files with no extension. Specifying two consecutive ** asterisk characters includes all files regardless of extension.

-F, --format=FORMAT

Defines in which format output should be generated; the default format is *text* and the available formats are: html,htmlembedded,json (see [OUTPUT FORMATS](#))

--grading[=BOOL]

Show statistics and information in a way that is formatted for grading of student projects; this is the same as supplying the options **-HlmrTw**

-H, --hard[=BOOL]

Track lines and look for duplicates harder; this can be quite slow with big repositories

-l, --list-file-types[=BOOL]

List all the file extensions available in the current branch of the repository

-L, --localize-output[=BOOL]

Localize the generated output to the selected system language if a translation is available

-m, --metrics[=BOOL]

Include checks for certain metrics during the analysis of commits

-M, --forcemonths[=BOOL]

Show statistics in months as well as in weeks when **-w** is used. The blame output then gets both a monthly and a weekly age column, and the timeline is printed both monthly and weekly

-r, --responsibilities[=BOOL]

Show which files the different authors seem most responsible for

--since=DATE

Only show statistics for commits more recent than a specific date

-T, --timeline[=BOOL]

Show commit timeline, including author names

--until=DATE

Only show statistics for commits made up to a specific date

-w, --weeks[=BOOL]

Show all statistical information in weeks instead of in months

-x, --exclude=PATTERN

An exclusion pattern describing the file paths, revisions, revisions with certain commit messages, author names or author emails that should be excluded from the statistics; can be specified multiple times (see [FILTERING](#))

-h, --help

Display help and exit

--version

Output version information and exit

OUTPUT FORMATS

There is support for multiple output formats in gitinspector. They can be selected using the **-F/--format** flags when running the main gitinspector script.

text (plain text)

Plain text with some very simple ANSI formatting, suitable for console output. This is the format chosen by default in gitinspector.

html

HTML with external links. The generated page has a responsive layout that adapts to phones and tablets, a light and a dark mode, sortable tables and a search box, and it can be opened in a given view straight from the link (see [VIEW PARAMETERS](#)). It loads the report font from Google Fonts and the author avatars from Gravatar, and therefore needs an active internet connection to look the way it is meant to.

htmlembedded

HTML with no external links. Similar to the HTML output format, but requires no active internet connection; the font falls back to the one installed on the machine and the avatars are drawn from the initials of each author. As a consequence; the generated pages are somewhat bigger, as the stylesheet and the script have to be embedded into the generated output.

json

JSON suitable for machine consumption. If you want to parse the output generated by gitinspector in a script or application of your own; this format is suitable.

xml

XML suitable for machine consumption. If you want to parse the output generated by gitinspector in a script or application of your own; this format is suitable.

VIEW PARAMETERS

The HTML reports remember how the reader left them; the theme, the light or dark mode and whether the minor authors are folded away. A link can pick that view up front by adding parameters to the address:

- **report.html?theme=terminal**, open the report in the terminal theme; the themes are *default* and *terminal*
- **report.html?mode=dark**, open the report in dark mode; the modes are *light* and *dark*

- **report.html?minor=shown**, open the report with the minor authors listed; the values are *shown* and *hidden*, with *true/false* and *yes/no* accepted just as well

The parameters can be combined (**report.html?theme=terminal&mode=dark&minor=shown**) and a value outside the listed ones is ignored. They only decide how the report opens; whatever the reader picks afterwards is remembered as usual and is never overwritten by a link.

FILTERING

gitinspector offers several different ways of filtering out unwanted information from the generated statistics:

- **gitinspector -x myfile**, filter out and exclude statistics from all files (or paths) with the string "myfile"
- **gitinspector -x file:myfile**, filter out and exclude statistics from all files (or paths) with the string "myfile"
- **gitinspector -x author:John**, filter out and exclude statistics from all authors containing the string "John"
- **gitinspector -x email:@gmail.com**, filter out and exclude statistics from all authors with a gmail account
- **gitinspector -x revision:8755fb33**, filter out and exclude statistics from all revisions containing the hash "8755fb33"
- **gitinspector -x message:BUGFIX**, filter out and exclude statistics from all revisions containing "BUGFIX" in the commit message.

The gitinspector command also lets you add multiple filtering rules by simply specifying the -x options several times or by separating each filtering rule with a comma;

- **gitinspector -x author:John -x email:@gmail.com**
- **gitinspector -x author:John,email:@gmail.com**

Sometimes, sub-string matching (as described above) is simply not enough. Therefore, gitinspector lets you specify regular expressions as filtering rules. This makes filtering much more flexible:

- **gitinspector -x "author:^(?!John Smith)\$"**, only show statistics from author "John Smith"
- **gitinspector -x "author:^(?![A-C])"**, only show statistics from authors starting with the letters A/B/C
- **gitinspector -x "email:.com\$"**, filter out statistics from all email addresses ending with ".com"

The same works for file paths, which limits the statistics to one or more sub directories of a repository:

- **gitinspector -x "^(?!subdir/)"**, only show statistics from files in the directory "subdir"
- **gitinspector -x "^(?!src/lib/)"**, only show statistics from files in the directories "src" and "lib"

USING GIT TO CONFIGURE GITINSPECTOR

Options in gitinspector can be set using **git config**. Consequently, it is possible to configure gitinspector behavior globally (in all git repositories) or locally (in a specific git repository). It also means that settings will be permanently stored. All the long options that can be given to gitinspector can also be configured via git config (and take the same arguments).

To configure how gitinspector should behave in all git repositories, execute the following git command:

git config --global inspector.option setting

To configure how gitinspector should behave in a specific git repository, execute the following git command (with the current directory standing inside the repository in question):

git config inspector.option setting

REQUIREMENTS

gitinspector runs on Python 2.7 as well as on Python 3.5 and later, and needs the **git** executable in the PATH. The exact versions the project keeps working on are listed in the REQUIREMENTS.txt file shipped with the documentation.

AUTHOR

Originally written by Adam Waldenberg.

REPORTING BUGS

Report gitinspector bugs to gitinspector@ejwa.se

The gitinspector project page: <https://github.com/ejwa/gitinspector>

If you encounter problems, be sure to read the FAQ first: <https://github.com/ejwa/gitinspector/wiki/FAQ>

There is also an issue tracker at: <https://github.com/ejwa/gitinspector/issues>

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SEE ALSO

git(1)